The Simple Linear Reg Model

ideal, population Level model

$$y_t = \beta_0 + \beta_1 x_{t,1} + \varepsilon_t$$

y_t : observed value for my response variable (target, outcome, dependant) at time t

β_0 : expected value (population mean) of y when $x_{t,1} = 0$

$x_{t,1}$: observed value of our predictor (independant) var at time t

ε_t : random error term

assume that these $\varepsilon_t \sim$ iid $N(0, \sigma^2)$

The Multiple Linear Regression Model

>1 predictors $(2, 3, \ldots, q)$

β_0 : expected value of y when $\boxed{\text{all}}$ predictors are set to 0.

β_i : for one unit increase in predictor i while keeping all other pred constant, β_i represent the expected change in y_t